

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 04

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Food: cod, tuna, swordfish, sharks, turtles; extinct - dodo, great auk, passenger pigeon (1) • Fashion: leopard, snow leopard, tiger, ocelot fur; fur seals; crocodile / alligator leather; kingfisher, parrot, ostrich feathers (1) • Entertainment / pets: parrots, lizards, snakes, tortoises, tropical marine fish (most wild-caught), dolphins / orcas for marine parks, wild-collected air plants (1) • Furniture / ornaments: mahogany, teak, ramin timber; ebony and ivory piano keys; turtle shell, shark teeth, coral and shells as souvenirs (1) • Traditional medicines: tiger (bones, claws), rhinoceros horn, seahorses (1)	5
01.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Threat to humans: sharks, venomous snakes, crocodiles (1) • Disease vectors: malaria mosquitoes, tsetse flies (1) • Predators of livestock: wolves, puma, birds of prey, herons, seals (1) • Agricultural / forestry pests and competitors: rabbits, deer, wood-boring beetles, squirrels, beavers (1)	3
01.3	<i>1 mark for each point.</i> • High demand and prices make illegal trade / poaching profitable; laws are hard to enforce across borders (1) • Belief in the medicine persists despite little scientific evidence; species become rarer and more valuable (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-21; AQA spec 3.1.2.2.	

Question	Marking guidance	Marks
02.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • By-catch = $5.9 + 1.9 = 7.8$ t; total = 8.4 t (1) • $7.8 / 8.4 \times 100 = 92.9\%$ (accept 93%) (1)	2
02.2	<i>1 mark.</i> • The nets have a very small mesh size so few non-target animals can escape (1)	1
02.3	<i>1 mark for each point.</i> • They breed at an old age (skate about 11 years; Greenland shark 100-150 years) and produce few young (1) • They may reach a catchable size before they are sexually mature, so fishing can remove the whole breeding population before it has bred (1) • Populations recover slowly, unlike r-selected species (herring, mackerel) which breed young and produce many young; they are caught as by-catch in demersal trawls so fishing continues even though they have no commercial value (1)	3
02.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Impacts: population decline of k-selected species, by-catch (immature target fish, over-quota fish, non-commercial species; drift nets - whales, dolphins, turtles; longlines - albatrosses; pair trawls - porpoises), ghost fishing, seabed damage by demersal trawls, coral and seagrass damage, dynamite fishing, food-web effects • Catch quotas: limit weight landed; work well for single-species pelagic shoals but in mixed fisheries over-quota fish are discarded dead so the quota fails to protect them • Gear design: mesh size and square-mesh escape panels let small fish escape (diagonal mesh closes under drag); turtle exclusion devices; acoustic pingers for dolphins; curved hooks, decoys, sinkers and night setting to protect albatrosses; biodegradable / radio-tracked traps to reduce ghost fishing • Restricted effort: limits on boat size, engine power and days at sea; decommissioning; restricted methods - drift-net bans, demersal trawling bans on sensitive seabeds (Rockall Bank, Hatton Bank) • Evaluation: enforcement at sea is difficult; some methods reduce catch or raise costs; quotas need reliable stock data; effectiveness depends on species (single vs mixed shoals) and on international cooperation as fish migrate; as local stocks decline fishing moves to more distant waters • Justified conclusion weighing effectiveness of different methods	9

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 341-345; AQA spec 3.5.2.2.	
03.1	<i>1 mark for each point.</i> • Images / maps of large areas; land-use change, deforestation, flooding (1) • Vegetation density and surface temperature (1) • Wind velocity, sea-surface altitude, wave height, oil pollution (1)	3
03.2	<i>1 mark.</i> • Microwaves pass through clouds (unlike visible light) (1)	1
03.3	<i>1 mark for each point.</i> • Very large / remote areas can be surveyed quickly and repeatedly so changes over time are detected (1) • No need for researchers to visit inaccessible areas, reducing cost and disturbance; data are standardised across the whole area (1)	2
03.4	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Motion-sensitive cameras placed along trails record animals passing, using visible light by day and infrared at night (1) • Each tiger has a unique pattern of facial / body stripes (1) • Images are compared with the database so individuals are recognised rather than counted twice (1) • The number of different individuals recorded gives a minimum population; repeated captures give territory size, movements, lifespans and social groupings (1) • Limitation: only works for species with unique visible features (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-403, 406-407; AQA spec 3.1.2.3.6.	
04.1	<i>1 mark for the correct answer.</i> • $(240\,000 - 2\,000) / 240\,000 \times 100 = 99.2\%$ (1)	1
04.2	<i>1 mark for each point.</i> • Whales are k-selected: slow growing, late maturing, one calf every few years, so populations recover very slowly (1) • Technology (harpoon cannons, steam / factory ships) allowed catches far above the sustainable yield within decades (1) • No regulation / open access to the ocean; whales were valuable (oil, spermaceti) and easily found (1)	3
04.3	<i>1 mark for each point.</i> • Moratorium on commercial whaling since 1986 until populations recover (1) • Total protection for certain species; whale sanctuaries (1) • Limits on numbers and sizes taken; protection of suckling mothers and calves (1) • Research into whale biology and threats (ship strikes, net entanglement); exceptions: aboriginal subsistence quotas (eg Inuit bowheads), 'scientific' permits (Japan - ruled not research in 2014), Norway / Iceland set their own quotas (1)	4

Question	Marking guidance	Marks
04.4	<i>1 mark for each point.</i> • ITTO's 1990 target of trade only from sustainably managed forests by 2000 was not met and little progress since (1) • Timber comes from many countries / private land with weak enforcement, high demand and alternative land uses; whaling involves few nations and a single treaty body (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 31-32; AQA spec 3.1.2.3.2.	
05.1	<i>1 mark.</i> • Development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs (1)	1
05.2	<i>1 mark for each point.</i> • Impact depends on total population x per capita impact (1) • With few people, natural processes (forest regrowth, fish recruitment, aerobic decomposition / dilution) recover faster than the resource is used or the waste added (1) • Larger populations exceed the rate of natural recovery: forest and soil do not regenerate, catches exceed MSY, water deoxygenates (1)	3
05.3	<i>1 mark for each point.</i> • Urban areas are supported by resources brought in from elsewhere; when one area is depleted, exploitation simply moves (timber, fish, minerals, energy) so consumers see no shortage (1) • Rural communities exploit local resources directly so the effects of over-exploitation become obvious quickly, encouraging cultural limits to need (1)	2
05.4	<i>Up to 4 marks in total; award marks in proportion to the completeness of each point.</i> • Climate: warming increases evaporation, cloud and albedo; higher CO₂ increases photosynthesis and sequestration; hydrological cycle: more evaporation gives more precipitation; populations: density-dependent factors return numbers to carrying capacity (up to 2) • In agriculture soil degradation, water shortage and loss of predators would reduce yields (negative feedback limiting the activity) (up to 2) • Humans respond with fertilisers, irrigation and pesticides to maintain yields, further damaging the natural systems that support production (up to 2)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-372; AQA spec 3.6.1.	
06.1	<i>Correct answer only.</i> • B (1)	1
06.2	<i>1 mark for each point.</i> • Rainforest: high light, warm, wet all year, stable conditions for long periods, continuous productivity, many niches (stratification) (1) • Antarctica: very low temperatures, low precipitation, extreme seasonal light, 98% ice cover so almost no terrestrial life (1) • Antarctic biodiversity is marine, supported by upwelling nutrients and the summer bloom; rainforest supports over half of terrestrial species on 10% of land (1) • Both are important carbon stores / climate regulators but rainforest species are specialists vulnerable to small changes; Antarctic species adapted to extreme fluctuation (1)	4

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-71; AQA spec 3.1.2.3.4.	
07.1	<i>1 mark for each point.</i> • Less dead organic matter / leaf litter so less humus and slower soil formation; fewer nutrients returned (1) • Loss of canopy and litter exposes soil to rain splash and wind (1) • Decomposing roots no longer bind soil so erosion, gullyng and landslides increase (especially on slopes) (1) • Reduced organic matter and biota lower infiltration and water retention (1)	4
07.2	<i>1 mark for each point.</i> • Reduced interception and transpiration so less water returned to the atmosphere and less rainfall downwind (1) • Increased runoff: more flooding after rain, lower flows in dry periods; sedimentation of rivers (1) • Increased albedo of cleared land (reflects more sunlight) (1) • Reduced carbon sequestration and loss of the carbon reservoir (CO₂ released by burning / decay) contributing to climate change (1)	4
07.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Fragmentation of remaining forest into isolated patches with non-viable populations / edge effects (1) • Loss of inter-species relationships (pollinators, seed dispersers) causing further declines (1) • Loss of genetic resources / potential medicines before they are discovered (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 368-369; AQA spec 3.5.3.3.	
08.1	<i>Correct answer only.</i> • Waste products that are the raw materials for other processes (1)	1
08.2	<i>1 mark for each point.</i> • Bacteria fix nitrogen at low temperature using enzymes and energy ultimately from sunlight (renewable) (1) • The Haber process needs high temperature and pressure powered by non-renewable fossil fuels, releasing CO₂ (1)	2
08.3	<i>1 mark for each point.</i> • Incineration uses high temperatures / fuel and releases pollutants; nutrients are not returned as usable raw materials (1) • Decomposition at low temperature recycles nutrients as the raw materials for other natural processes (material cycle) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2-3.6.3.	
09.1	<i>1 mark.</i> • Cross-bred offspring are healthier / more productive because the problems caused by homozygous recessive genes in inbred varieties are reduced (heterosis) (1)	1
09.2	<i>1 mark for each point.</i> • Zebu tolerate heat but have a low milk yield; Ayrshire have a high milk yield but are not heat tolerant (1) • Cross-breeding combined the desirable characteristics: heat tolerance with a high milk yield (1)	2

Question	Marking guidance	Marks
09.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Replacement of valuable animals that die with genetically identical individuals (1) • Replacement of herds culled during a disease outbreak (1) • Production of large numbers of individuals with desirable characteristics (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-322; AQA spec 3.5.1.2.	
10.1	<i>1 mark for each point.</i> • Plants produce chemicals (often alkaloids) that are toxic to the herbivores that would eat them (1) • In carefully controlled amounts these chemicals have beneficial physiological effects in humans (1)	2
10.2	<i>1 mark for each point.</i> • Only a very small proportion of species have been studied for medicinal substances, so it is not known which species will be useful (1) • Conserving the habitat protects all the species in it, including those not yet studied and the species they depend on (1)	2
10.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Marsupials (kangaroo / wallaby): young develop in the pouch so are easier to study than a human foetus - developmental problems (1) • Hippopotamus: hipposudoric acid skin secretion (sunscreen / antimicrobial) - burns treatment (1) • Marine sponges: proteins preventing graft rejection - organ transplants (1)	2

Question	Marking guidance	Marks
10.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Resources: wood, fibres, oils, fuels, food; new food species adapted to local conditions (morama bean, yeheb, perennial maize, Kernza, bison, eland) • Biomimetics: whale flipper tubercles, shark skin, termite-mound ventilation, bird bones, gecko adhesion, burrs / Velcro, lotus self-cleaning, spider silk • Medicines and physiological research; genetic resources - crop wild relatives, Vavilov centres, gene pools, inbreeding • Ecosystem services: atmospheric composition, hydrological cycle, biogeochemical cycles, soil maintenance; interspecies relationships - food, pollination, seed dispersal, habitat provision • Evaluation: services go unnoticed because of dynamic equilibrium; unknown future value; economic vs moral / intrinsic arguments; conservation also needed for aesthetic and cultural reasons • Justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	

Question	Marking guidance	Marks
11.1	Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) - Physical features: mass / gravity retaining an atmosphere and giving pressure for liquid water; distance from the Sun and the Goldilocks zone; speed of rotation reducing temperature extremes; tilt producing seasons; magnetic field deflecting the solar wind; the Moon stabilising the orbit and producing tides - Importance of liquid water: solvent, transport, temperature control, anomalous expansion, high specific heat capacity, habitats, absorption of UV before the ozone layer - Development of life: Archaea at vents; photosynthesis; oxygen reacting with iron then accumulating; ozone layer allowing colonisation of land; aerobic organisms - Life changing conditions: carbon sequestration offsetting solar brightening, biogeochemical cycles, transpiration and rainfall, regulation of atmospheric gases - Methods of studying past conditions: proxy data, isotope ratios, ice cores, Mg:Ca ratios; understanding of continental drift and ocean currents - Links to current threats (climate change, ozone depletion) and a justified conclusion about the interdependence of physical and biological factors Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-9; AQA spec 3.1.1.	25

Question	Marking guidance	Marks
11.2	Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Resources: wood, fibres, oils, fuels, food; new food species (morama bean, yeheb, perennial maize, Kernza, bison, eland) and their adaptations • Biomimetics: wing-tip feathers, whale tubercles, shark skin, termite mounds, bird bones, gecko adhesion, burrs, lotus, spider silk • Medicines: poppies, cinchona, willow, yew (taxol), Mexican yam, sponges (AZT), sharks; few species yet studied; physiological research (marsupials, hippo, sponges) • Genetic resources: crop wild relatives, Vavilov centres, gene pools and inbreeding • Ecosystem services: atmospheric regulation, hydrological cycle, biogeochemical cycles, soil maintenance; interspecies relationships - food, pollination (Darwin's orchid), seed dispersal, habitat provision • Evaluation: economic vs intrinsic / moral arguments; services unnoticed because of dynamic equilibrium; unknown future value; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	25